Combined Question – Snowpipe and Dynamic Table Transformations (Employee Dataset)

Marks: 60 | Difficulty: Medium

Objective

Build an end-to-end Snowflake data pipeline using the Employee dataset from the `Employee_SCD_Dynamic` folder. Ingest employee records from AWS S3 using Snowpipe, monitor ingestion status, and create Dynamic Tables for employee aggregation, SCD Type 1, and SCD Type 2 processing.

Dataset

- `employee_batch1.csv` — 18 rows
- `employee_batch2.csv` — 8 rows
- `employee_batch3.csv` — 2 rows
- **Expected total: 28 raw records and 20 unique employees**

Step 1 – Create the Environment (10 Marks)

Create the database `EMPLOYEE_SCD_DB`, schema `RAW`, CSV file format `CSV_FMT`, external stage `S3_EMP_STAGE`, and raw table `EMPLOYEE_RAW`.

Create the raw table with these columns:

```
EMP_ID, EMP_NAME, DEPARTMENT, DESIGNATION, WORK_LOCATION, UPDATED_AT,
FILE_NAME, LOADED_AT
```

Configure the external stage to access the AWS S3 dataset folder.

Step 2 – Create Snowpipe and Ingest Data (10 Marks)

Create `EMPLOYEE_PIPE` with `AUTO_INGEST = TRUE` to load employee records from the external stage into `EMPLOYEE_RAW`.

- Ingest all three batches.
- Populate `FILE_NAME` using `METADATA$FILENAME`.
- Verify 18 rows after Batch 1, 26 after Batch 2, and 28 after Batch 3.
- Group the raw records by `FILE_NAME` to validate all files.

Step 3 – Monitor Snowpipe (5 Marks)

Use `SYSTEM$PIPE_STATUS` and `INFORMATION_SCHEMA.COPY_HISTORY` to verify the pipe status, successful file ingestion, and any loading errors.

Submit the SQL queries and their results as evidence.

Step 4 – Create Employee Aggregation Dynamic Table (10 Marks)

Create `DT_EMPLOYEE_DEPT_SUMMARY` with `TARGET_LAG = '1 minute'`.

Aggregate employee data by `DEPARTMENT` and calculate:

- `TOTAL_EMPLOYEES`
- `TOTAL_LOCATIONS`
- `LATEST_UPDATE`

Validate the departmental counts and confirm that the aggregation reflects the ingested records.

Step 5 – Create SCD Type 1 Dynamic Table (10 Marks)

Create `DT_EMPLOYEE_SCD1` with `TARGET_LAG = '1 minute'`.

Use `ROW_NUMBER()` partitioned by `EMP_ID` and ordered by `UPDATED_AT DESC` to retain only the latest record for each employee.

Required columns:

```
EMP_ID, EMP_NAME, DEPARTMENT, DESIGNATION, WORK_LOCATION, LAST_UPDATED_AT
```

Validate the following:

- Employee 6102 (Meera Iyer) shows `DESIGNATION = 'Lead Data Engineer'` and `WORK_LOCATION = 'Bengaluru'`.
- Employee 6108 (Priya Menon) shows `WORK_LOCATION = 'Mumbai'`.
- Employees 6119 and 6120 appear in the final output.
- Expected output: **20 unique employees**.

Step 6 – Create SCD Type 2 Dynamic Table (10 Marks)

Create `DT_EMPLOYEE_SCD2` with `TARGET_LAG = '1 minute'` to retain employee change history.

Required columns:

```
EMP_ID, EMP_NAME, DEPARTMENT, DESIGNATION, WORK_LOCATION, EFFECTIVE_FROM,
EFFECTIVE_TO, IS_CURRENT, VERSION_NO
```

Use `ROW_NUMBER()` and `LEAD()` over each employee's update history to derive version numbers, effective dates, and current-record flags.

Validate that:

- Each employee's versions are ordered by `UPDATED_AT`.
- Previous versions have `IS_CURRENT = FALSE`.

- The latest version has `IS_CURRENT = TRUE` and `EFFECTIVE_TO = NULL`.
- Employees with no changes have one version.
- All 28 raw records are represented in the resulting history, assuming each source row represents a distinct employee version.

Step 7 – Refresh, Validate, and Pause the Pipe (5 Marks)

Verify that all three Dynamic Tables have refreshed successfully and validate their output.

Finally, pause `EMPLOYEE_PIPE` and confirm its final status.

Evaluation Rubric

No.	Criterion	Marks
1	Environment setup and Snowpipe ingestion	10
2	Pipe monitoring and file validation	5
3	Employee aggregation Dynamic Table	10
4	SCD Type 1 Dynamic Table	10
5	SCD Type 2 Dynamic Table	10
6	Refresh verification and final pipe state	5
7	Key-record validation and submission evidence	10
	Total	60